

AEGIS R1 / R2 / R3

Three-Runner Implementation & Strategy Document

Formulas, pseudocode, validation methodology, and an original cross-runner strategy for running R1, R2, and R3 as genuinely complementary, fully isolated parallel engines — with a direct answer to keeping daily recommendation throughput alive.

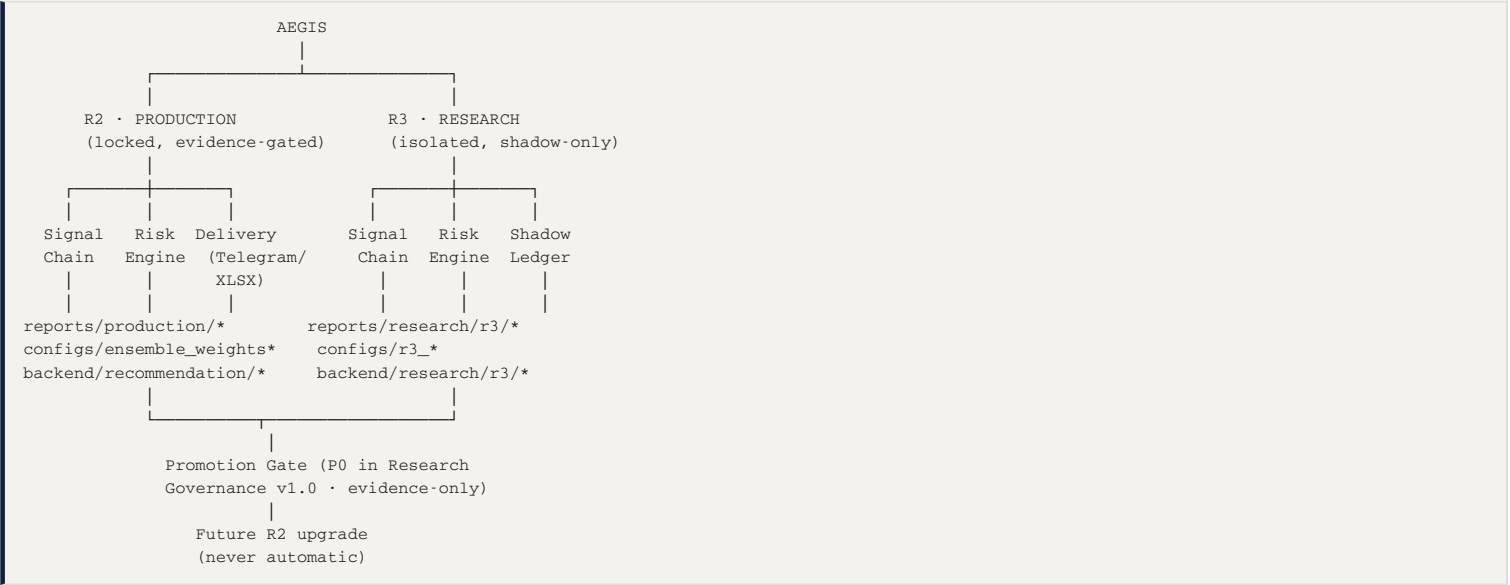
Scope	(1) R1 self-analysis, real historical performance, and a cross-runner composite strategy, (2) implementation-grade detail for the locked R2 upgrade list, (3) deeper technique-level specification for the Runner 3 research catalog, (4) a formal isolation contract keeping all three separated, (5) validation methodology and a phased rollout plan
Ground rules carried forward	R2 stays locked production — every upgrade here is evidence-gated, not a direct edit. R1 stays retired from production P&L, engine-alive for research/advisory only. R3 is a fully isolated research challenger. All three are structurally isolated from each other — no shared write paths, no shared model artifacts, no silent promotion between them.
Live operational finding	R2 has produced zero new positions in 28 days; R1 (still engine-alive though retired from delivery) produced 11 live picks the same day this analysis was done. Diagnosing R2's silence is treated as more urgent than exposing R1's output — see Part R1.
Prepared	2026-09-02

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Part 0 · System Architecture & the R1/R2/R3 Isolation Contract

Every upgrade in this document assumes the following architecture, stated explicitly so it can be enforced mechanically rather than by convention:



Isolation guarantees (mechanically enforced, not just documented)

Guarantee	Mechanism	Precedent
R3 never writes to a path R2 reads	All R3 outputs live under <code>reports/research/r3/</code> and <code>configs/r3_*.yaml</code> — disjoint from every path R2's daily driver reads. A CI check (<code>test_r3_no_production_writes</code>) fails the build if any R3 module imports or writes a production path.	Same pattern as the R1 retirement's <code>overrideallow</code> invariant — a mechanical check, not a promise.
R3 never shares model artifacts with R2	R3 trains its own model files under a separate namespace; no shared pickle/weights file is ever loaded by both runners. R2's adaptive ensemble weights (<code>configs/ensemble_weights_adaptive.yaml</code>) remain untouched by any R3 process.	Consistent with the "no <code>overrideallow</code> anywhere" invariant already verified in the 2026-09-01 closure.
R3 never appears in the delivered workbook	Same contract already built for R1 retirement (<code>backend/delivery/canonical/retirement.py</code>) extends naturally: R3 is <code>SHADOW_ONLY</code> by default in <code>configs/aegis_runner_registry.yaml</code> until a promotion decision explicitly flips it.	Reuses the exact filter logic that currently keeps retired R1 out of Portfolio/Exit History/Monthly Summary.
R3 has its own Position ID namespace	Using the canonical <code>make_opportunity_id(market, runner, ticker, created_date)</code> authority from Phase 2 — <code>runner="R3"</code> produces PIDs that are structurally distinct and never collide with R1/R2 PIDs, and R3 positions never enter the R2 Registry.	Direct reuse of the Phase 2 Identity Migration's canonical PID authority — no new ID scheme invented.
R1's advisory/composite output never touches R2 production paths	R1 stays engine-alive but delivery-suppressed per the existing retirement contract; any composite-layer output (Part R1, \$R1.5) is a new, separate artifact that reads R1/R2/R3 outputs but writes to none of them — a read-only aggregator, not a fourth peer with write access to anyone's state.	Extends the retirement filter logic (<code>backend/delivery/canonical/retirement.py</code>) to a three-way (soon four-way, if a composite layer ships) isolation model.
Promotion is a human decision, never automatic	Even after R3 clears every statistical gate in Part 4, moving anything from R3 into R2 requires the same CEO authorization pattern already used for R1 retirement and Phase 2 — a named, dated, written authorization, not a merge.	Matches "GO FINAL LOCK" / "GO P2 · DOCUMENT + EXECUTE" precedent.

This is what "parallel without disturbing or mixing" means concretely: not two teams being careful, but two runners that are structurally incapable of writing into each other's state, verified by the same class of CI gate that already protects R2 from R1's retirement.

Part R1 · R1 Engine — Self-Analysis, Real Performance, and the Three-Runner Strategy

This part does four things: verifies what R1 actually is (not assumed — traced to code), reports its real historical performance including the sector/filter analysis that triggered this section, gives an independent critical read on the proposed exposure plan, and lays out an original synthesis for how three parallel runners can genuinely complement each other rather than just coexist.

R1.1 · What R1 Actually Is (verified)

Traced to `research/adaptive_rec_v2/`, `data/aegis_today.csv`, and `configs/aegis_retirement.yaml`: R1 fits three candidate models daily (baseline heuristic, HistGradientBoosting v2.0, Logistic Regression v2.0), selects the winner by a Precision@10 top-K rule on rolling history, produces calibrated confidence, and outputs a profile-based verdict (Shield/Conservative, Growth) with explicit regime-conditional confidence adjustment ("regime cautious → partial deploy"). The retirement was delivery-layer suppression only — per the config's own comment, engine-level disable "requires explicit code modification... not covered by this delivery-layer retirement flag." R1 is verifiably still running and still producing daily output.

R1.2 · R1's Real Historical Performance (25 genuine closed trades)

Pulled directly from the Registry, excluding administrative exits, both markets:

Market	n	Win Rate	Mean P&L	Median P&L	Read
India	4	0.0%	−5.48%	−6.26%	4-for-4 losses, all closed within days of each other — too small and too clustered to conclude India R1 is structurally broken, but a real red flag worth re-validating before use.
USA	21	61.9%	+1.18%	+0.33%	Genuinely positive, but this is n=21, "research signal" tier at best per the locked sample rules, not "validation candidate."
Combined	25	52.0%	+0.11%	—	Barely breaking even blended — India's losses are dragging an otherwise-positive USA book down to roughly flat.

SECTOR BREAKDOWN — WHERE THE SIGNAL CONCENTRATES

Sector	n	Win Rate	Mean P&L	Tickers
Materials	1	100%	+6.94%	CF
Energy	1	100%	+4.73%	APA
Healthcare	5	80%	+2.23%	SUNPHARMA, DGX, LH, BDX, AMGN
Technology	3	67%	+1.01%	ADSK, AAPL, NVDA
Financials	6	50%	+0.04%	TRV, WTW, V, GS, CPAY, JPM
Consumer Staples	1	0%	−0.18%	KO
Consumer Discretionary	2	50%	−0.34%	ROST, TGT
Industrials	3	33%	−0.88%	HON, RTX, MMM
Utilities	2	0%	−6.26%	POWERGRID, NTPC
Consumer Defensive	1	0%	−7.21%	ITC

FILTER-STRATEGY HYPOTHETICALS (SAME 25 TRADES, TESTED AGAINST THE "SKIP THE LOSERS" QUESTION)

Strategy	n	Win Rate	Mean	Sum	vs. baseline
Baseline (all 25)	25	52%	+0.11%	+2.84%	1.0×
Skip losing sectors	17	65%	+1.52%	+25.89%	9.1×
Skip India entirely	21	62%	+1.18%	+24.77%	8.7×
Hard −3% stop	25	—	+0.55%	+13.68%	4.8×
Winning-sectors only	16	69%	+1.63%	+26.07%	9.2×

The 9× number is real arithmetic on this sample, and it is also a textbook overfitting trap if treated as a rule to ship. Three things make this explicit: (1) n=25 is "research signal" tier, one tier below what this project's own locked rules consider strong evidence; (2) nearly all 25 trades closed within a single narrow window (most in late August 2026) — this is one regime's worth of data, not a tested pattern; (3) the losing sectors (Utilities, Consumer Defensive, Consumer Staples) are classically *defensive* sectors — in a risk-off regime, exactly the opposite ranking would likely hold. A hardcoded "skip Utilities" rule fitted to August 2026 could actively hurt performance the next time the market turns risk-off, which is precisely when defensive sectors are supposed to earn their keep.

R1.3 · My Independent Read on the Exposure Plan

Three points, building on but not simply restating the proposal already on the table:

- Sequencing:** shipping an R1 advisory sheet on a fixed calendar date, with the R2 zero-entry diagnosis running "in parallel," has the priority backwards. If R2's 28-day silence is a bug, R1-as-stopgap is genuinely useful. If it's R2 correctly sitting out a bad regime (this project's own data: WEAKENING-regime trades run a 28.6% win rate, 2.29× downside capture), then showing the operator 11 confident BUY picks from a documented weaker engine at exactly the moment the stronger engine is correctly silent risks the operator overriding good discipline with worse information. Gate the advisory ship date on at least a preliminary read of *why* R2 is silent.
- Governance gap:** none of the seven locked Research Triggers catch "a runner went silent." Trigger 4 (Subsystem Failure) assumes bad output, not absent output. Closed formally in §R1.7 below.
- Unquantified risk in "zero coupling, zero risk":** R1 was never wired to the dynamic exit bridge (P0 in the R2 upgrade list) — it was

retired before that bridge existed in enforced form. If an operator informally acts on an "advisory" R1 pick, that position sits completely outside every exit-discipline mechanism this project has spent most of its effort building. "Not counted in production P&L" is correct for reporting, but doesn't mean a human can't still take real, ungoverned risk off the advisory feed. State this explicitly in the operator-facing banner, not just "advisory only."

R1.4 · Path A vs. Path B — Where R1's Value Actually Lives

Path A (fold R1 into R2's ensemble as model #12) is the wrong lever, for reasons the evidence itself confirms: R2's ensemble isn't currently the bottleneck — its acceptance gates are (0 new picks in 28 days despite 11 models scoring every day). Adding a 12th input feeds more signal into a pipeline that's discarding all signal downstream. Worse, R1 and R2 already look at heavily overlapping information (momentum, trend, quality, sector, news, regime) — ensemble theory says a correlated 12th member adds little, and IC-based weighting would demote R1's mediocre standalone signal (52% WR, +0.11% mean) to near-zero influence within weeks, quietly erasing the one thing R1 has that R2 doesn't: a coherent, opinionated conservative profile (Shield). Melting that into an averaged ensemble destroys it rather than using it.

Path B (R1 stays a separate, independent leg, feeding a composite layer that reads both) is the right direction — but the version proposed so far (a 4-bucket rule table: HIGH CONVICTION / R1 CONVICTION / R2 CONVICTION / CONFLICT) is a reasonable first cut that can be made more principled. That's the subject of §R1.5.

R1.5 · An Original Synthesis: The AEGIS Three-Runner Strategy

This is the deeper answer to "how do we make three parallel runners genuinely complementary, not just three copies of the same idea running side by side." Four ideas, each building on work already validated elsewhere in this project rather than inventing new machinery from scratch.

1 · DIFFERENTIATED MANDATES, NOT THREE GENERALISTS

Right now R1 and R2 are two attempts at answering the same question ("what's a good stock to buy") with overlapping information. Three runners running the same mandate is redundancy of opinion, not diversification of strategy. A sharper structure, grounded in what the evidence already shows each runner is naturally suited to:

Runner	Mandate	Why this runner, per the evidence
R1	Conservative / defensive specialist	Already has an explicit Shield/Conservative profile and regime-conditional confidence. Its USA book (61.9% WR) skews toward lower-volatility, larger names. Formalize this as R1's actual job, rather than "R2 but older."
R2	Broad all-weather core	The 11-model IC-weighted ensemble, genuinely comprehensive across technical/fundamental/macro/event/news. This is correctly the platform's primary engine — keep it there.
R3	Aggressive / innovative research edge	Small size, high evidence bar, house for the newer techniques (Knowledge-Graph scoring, factor-neutral construction, multi-horizon consensus) from Part R3. Explicitly the place where new ideas earn their way toward R2, not a third production copy.

This turns "three runners" from three redundant opinions into a genuine barbell: a defensive anchor, a broad core, and a research edge — each with its own evidence bar, and each useful specifically *because* it's different from the other two, not despite it.

2 · A META-ENSEMBLE COMPOSITE LAYER, USING R2'S OWN PROVEN MATH ONE LEVEL UP

Rather than a hand-built rule table (4 fixed buckets with fixed confidence numbers), reuse the same IC-adaptive weighting mechanism R2 already uses internally to blend its 11 models — applied one level higher, to blend across runners:

Composite_Score(stock) = Σ_r Trust_Weight(r) · Runner_Score_r(stock), for r ∈ {R1, R2, R3}

Trust_Weight(r) recalculated daily from each runner's trailing realized IC, exactly like R2's internal model weights (configs/ensemble_weights_adaptive.yaml) — same mechanism, new scope

```
def composite_score(stock, runner_scores, runner_trust_weights):
    # runner_trust_weights computed the same way as R2's internal
    # per-model IC weights — reuse compute_adaptive_weights(), just
    # called on {R1, R2, R3} trailing IC instead of {momentum, trend, ...}
    return sum(runner_trust_weights[r] * runner_scores[r].get(stock, 0)
               for r in ["R1", "R2", "R3"])

def cross_runner_conviction(stock, runner_scores):
    present = [r for r in ["R1","R2","R3"] if stock in runner_scores[r]]
    if len(present) >= 2:
        agreement = agreement_rate(runner_scores, present, stock)
        return {"n_runners": len(present), "agreement": agreement,
                "class": classify(len(present), agreement)}
    return {"n_runners": len(present), "agreement": None, "class": "SINGLE_RUNNER"}
```

This is more auditable than a fixed rule table (every weight is explainable the same way R2's model weights already are), self-consistent with the platform's existing math, and naturally extends to a future R4 without redesign.

3 · KNOWLEDGE-GRAPH-BASED ROLLING GROUP FILTER, REPLACING STATIC SECTOR LABELS

The proposed rolling sector-momentum filter (rank sectors by trailing realized P&L + news sentiment + relative strength + regime multiplier, only accept top-N sectors) is the right shape — regime-adaptive, not a hardcoded skip list. The original synthesis here: use the Knowledge Graph's **57 already-computed communities** (Part R3, §II) as the grouping unit instead of, or alongside, static GICS/sector labels.

Group_Composite_Score(g, asof) = 0.4·Trailing20d_Realized_PnL(g) + 0.3·Trailing10d_NewsSentiment(g)

+ 0.2·Trailing60d_RelativeStrength(g) + 0.1·RegimeMultiplier(g, current_regime)

where g = Knowledge-Graph community, not static Sector — weights walk-forward tuned, not hand-set

Why this matters beyond "finer-grained than Sector = Utilities": a static sector label is exactly what produces the overfitting trap in §R1.2 (skip Utilities forever because it lost in August). A KG community is a data-driven, emergent grouping that can (and should) reshuffle as relationships in

the market actually change — it's structurally less prone to becoming a permanent, stale skip-list, because it isn't a fixed taxonomy in the first place.

4 · MINIMUM VIABLE SIGNAL — SOLVING "WE NEED DAILY RECOMMENDATIONS" WITHOUT ABANDONING THE GATES

This is the most consequential point in this section. Every piece of this project's evidence-gated methodology — the P0 Outcome Dataset, the walk-forward validation, the confidence recalibration, the interaction analyses — depends on a continuous stream of new positions to learn from. When R2 goes silent for 28 days, it isn't only a live-P&L problem: it's 28 days of missing data for *every* downstream research process in this document. The forward-validation engine this whole project is built around cannot validate a runner that isn't producing anything to validate.

The three-runner architecture is the structural fix, if used deliberately: with R1, R2, and R3 running independently, the composite layer's daily throughput doesn't depend on any single runner staying active. A concrete policy:

```
def daily_signal_floor_check(runner_outputs, min_signals=3):
    total = sum(len(runner_outputs[r]) for r in ["R1","R2","R3"])
    if total >= min_signals:
        return {"status": "OK", "gate_state": "NORMAL"}
    # Graceful, PRE-REGISTERED degradation — not a silent gate change,
    # not a manual operator override in a panic
    relaxed = widen_acceptance_gate(step=1)    # e.g. top-decile -> top-quintile
    relaxed_outputs = rerun_with_gate(relaxed)
    return {"status": "GATE_RELAXED", "gate_state": relaxed,
            "reason": f"only {total} signals at normal gate — widened one step",
            "flagged_to_operator": True}    # never silent
```

The relaxation step itself must be walk-forward validated and pre-registered (per Part 4's methodology) before it's allowed to fire in production — this isn't "loosen the rules when nervous," it's a tested, bounded, transparently-flagged fallback that only exists *because* three independent runners give the platform somewhere to look before resorting to it.

R1.6 · Runner Combination Matrix (use cases across R1 × R2 × R3)

Combination	Interpretation	Position-sizing implication
All 3 agree	Maximum cross-runner conviction — three independently-trained, differently-mandated engines converge	Full size, within existing risk-engine caps — still gated by P0's dynamic exit discipline
R2 + R3 agree, R1 silent/neutral	Core engine and research edge concur; R1's conservative screen simply has nothing to say here	Full-to-near-full size — treat as the platform's primary conviction case
R1 + R2 agree, R3 silent	Defensive and core engines concur; R3 hasn't reached a view yet (expected often, since R3 has a higher evidence bar)	Full size — this is close to today's implicit "R2 alone" case, now with defensive confirmation
R1 + R3 agree, R2 silent/AVOID	Notable: the conservative screen and the research edge concur while the core engine demurs — worth flagging as a genuine edge case for manual review, not auto-sized	Half size, WATCH-to-BUY, human review recommended
Only R2 has a view	Today's default state — the core engine alone	Standard current sizing rules
Only R1 has a view, R2/R3 silent	Exactly the current live situation (R2 at 0 for 28 days, R1 producing 11 picks) — the Minimum Viable Signal case	Reduced size, R1's own stop/target logic applies, flagged as "single-runner, gate-relaxed" per §R1.5.4
2-of-3 disagree (CONFLICT)	Genuine signal disagreement — treat as information (something about this stock is regime-ambiguous), not noise to average away	WATCH only, no auto-size — the disagreement itself is the useful output here
All 3 silent on a name	No engine sees an opportunity	No position — not itself a signal, just absence of one
Regime-conditional reweighting	In RISK_OFF/WEAKENING, upweight R1's defensive mandate and downweight R2/R3's more aggressive picks in the composite score; invert in confirmed BULL regimes	Changes <code>Trust_Weight(r)</code> in §R1.5.2's formula, not a separate rule — same mechanism, regime as an additional input
KG-community-conditional weighting	If evidence (per Part R2, §P4/P3 interaction work) shows a runner has a demonstrated edge within specific Knowledge-Graph communities, weight that runner higher for stocks in those communities specifically	Requires the P4 interaction study and per-runner community backtests first — Tier 2, evidence-gated

R1.7 · New Governance Additions

Eighth Research Trigger — Signal Silence: fires if any runner (R1, R2, R3, or future) produces zero new qualifying positions for $N \geq 10$ trading days while either (a) its own trailing baseline rate of new positions per 10 days is materially higher, or (b) a comparator runner or the raw candidate universe shows eligible setups under the same gates. Requires at least one comparator observation before firing — cannot fire if all runners are silent simultaneously, since that could be a genuine market-wide absence of opportunity rather than a subsystem fault.

Standing operational rule — Minimum Viable Signal: the composite layer's daily throughput (per §R1.5.4) is now itself a monitored metric, separate from any single runner's win rate or expectancy. A day where total cross-runner signal count falls below the pre-registered floor triggers the walk-forward-validated gate relaxation, always flagged to the operator, never silent.

R1.8 · What I Would NOT Do

- Not hardcode a sector or community skip list off the 25-trade sample — the defensive-sector losses are plausibly regime-specific and could invert in a risk-off period.
- Not fold R1 into R2's ensemble as a 12th model (Path A) — wrong lever, would erase R1's one differentiating asset (its conservative profile) for near-zero marginal signal.
- Not let R1 (or eventually R3) output flow into Registry, Exit History, or production P&L outside the composite layer's own governed path.
- Not ship the composite layer to production before it clears the same walk-forward validation and deflated-Sharpe correction (Part 4) as everything else in this document — the 9× filter number in §R1.2 is a hypothesis worth testing, not a result worth shipping.
- Not treat "three parallel runners" as done once they're all running — the value only materializes once the composite layer and differentiated mandates in §R1.5 actually exist; three independently-running, undifferentiated runners is not meaningfully different from one.

R1.9 · Staged Rollout

Stage	Scope	Effort	Gate to proceed
1	R1 advisory sheet in the workbook, gated on a preliminary R2 zero-entry read (not a fixed calendar date)	2-3 hours	Explicit banner stating R1 positions carry no dynamic-exit protection; R2 diagnosis at least preliminarily underway
2	Cross-runner composite spike: build the meta-ensemble scoring (§R1.5.2) and KG-community filter (§R1.5.3), backtest against the 25 R1 trades + all R2 closed trades, out-of-sample walk-forward, deflated for the number of weight/threshold combinations tried	2-3 weeks	Out-of-sample expectancy improvement holds after deflation; sector/community filter tested across at least one regime transition, not just August 2026
3	Differentiated mandates (§R1.5.1) formalized; Minimum Viable Signal policy (§R1.5.4) implemented and pre-registered; composite layer ships as its own workbook section, still fully isolated from R2 production per Part 0's contract	ongoing	Signal Silence trigger and Minimum Viable Signal floor both live and tested before this stage is considered complete

Part R2 · R2 Upgrade — Implementation Guide

Each item follows the same structure: problem/evidence, technical approach, formula or pseudocode, validation plan, rollout gate, and acceptance criteria. Every upgrade here is additive or calibration-layer — none changes core signal-generation logic without first clearing the evidence gate in Part 4.

P0 · Dynamic Exit Bridge — Retrospective Validation

Evidence: 539 historical R2 closes, zero triggered by stop/target/horizon logic. The bridge (STEP 52) is now wired but unproven — zero positions qualified for enforcement on its first live day.

TECHNICAL APPROACH

Replay every historical position through the now-wired `dynamic_risk_v2` logic using only information available up to each day (no lookahead), and compare the counterfactual exit against the actual historical exit.

Dynamic Stop(t) = Entry - k · ATR14(t) (k typically 1.5-2.5, trailing upward on new highs)
Target(t) = Entry + m · ATR14(t) (m > k, e.g. 1.5-2× the stop distance)
Horizon exit: force close at day = Horizon(d) if neither stop nor target hit

```
for each historical position P in outcome_dataset (n=539):
  replay_state = {entry_price: P.entry, high_water: P.entry, day: 0}
  for day, price in P.daily_price_series:
    # walk-forward, no lookahead
    atr = compute_atr14(P.ticker, asof=day)      # point-in-time ATR only
    stop = replay_state.high_water - k * atr
    target = P.entry + m * atr
    replay_state.high_water = max(replay_state.high_water, price)
    if price <= stop:
      counterfactual_exit = ("STOP", day, price); break
    if price >= target:
      counterfactual_exit = ("TARGET", day, price); break
    if day >= P.horizon_days:
      counterfactual_exit = ("HORIZON", day, price); break
  else:
    counterfactual_exit = ("NONE_TRIGGERED", P.actual_exit_day, P.actual_exit_price)
  record(P.position_id, P.actual_exit_reason, P.actual_pnl,
         counterfactual_exit, counterfactual_pnl)
```

VALIDATION PLAN

- Compare distribution of actual P&L vs. counterfactual P&L across all 539 positions — paired bootstrap (10,000 resamples) for the mean difference, since returns are non-normal.
- Segment by regime (NORMAL/WEAKENING/RISK_OFF/CRASH/RECOVERY) — the hypothesis is that disciplined exits matter most in WEAKENING/RISK_OFF, where the platform's own data already shows 2.29× downside capture.
- Report: would-be-prevented max loss, would-be-forfeited max gain (discipline cuts both tails), and net expectancy delta.

LIVE MONITORING, ONCE VALIDATED

Expected fire rate ≈ P(price breaches k·ATR from high-water) — estimate from historical volatility distribution; alert if observed fire rate over trailing 20 trading days falls below 50% of this expectation.

Acceptance criteria: counterfactual expectancy ≥ actual historical expectancy at n≥50 (validation-candidate tier per the locked sample rules), with the live monitor showing at least one genuine enforcement fire within the first 20 trading days post-validation.

P1 · Confidence Calibration on Delivered Output

Evidence: DEV029 (Platt scaling, ECE 0.287–0.002) already proved raw confidence is miscalibrated at the research layer; this project's own India data reconfirmed near-zero-to-negative correlation between stated confidence and outcome twice more. The fix exists in research but was never wired to the number the operator actually sees in the Telegram message.

TECHNICAL APPROACH — TWO-STAGE CALIBRATION

Stage 1 (Platt scaling): P_calibrated = 1 / (1 + exp(A · raw_score + B))
fit A, B by maximum likelihood on held-out (raw_score, win/loss) pairs
Stage 2 (regime adjustment): P_final = P_calibrated · regime_multiplier(current_regime)
regime_multiplier estimated from realized win rate by regime bucket, clipped to [0.7, 1.3]

```
# Weekly recalibration job
def recalibrate_confidence(trailing_window_days=180):
  data = outcome_dataset.query(closed=True, window=trailing_window_days)
  if len(data) < 50:
    # validation-candidate floor
    return load_previous_calibration() # do not refit on too little data
  A, B = fit_platt(data.raw_confidence, data.win_loss)
  ece_before = expected_calibration_error(data.raw_confidence, data.win_loss)
  ece_after = expected_calibration_error(
    platt_transform(data.raw_confidence, A, B), data.win_loss)
  if ece_after >= ece_before:
    # sanity guard
    alert("recalibration made ECE worse — do not deploy")
    return load_previous_calibration()
  save_calibration(A, B, ece_after, asof=today())
  return (A, B)
```

VALIDATION PLAN

- Reliability diagram (predicted probability bucket vs. observed win rate), rebuilt weekly on a trailing 180-day window.
- Track Brier score and ECE over time as a first-class monitored metric, not a one-off research finding.
- Forward expectancy by calibrated-confidence decile, to confirm the calibrated number is actually monotonic with outcome (the raw number was not).

Acceptance criteria: ECE on trailing 180-day window ≤ 0.05 sustained over 4 consecutive weekly refits before the calibrated number replaces the raw one in the delivered Telegram/XLSX output. Confidence should **only** be allowed to influence position sizing after this gate clears — not before, per the locked "don't change sizing off an uncalibrated signal" rule.

P2 • Sector/Regime-Adjusted Ranking

Evidence: India read WEAKENING regime on 2026-09-01 with 28.6% win rate and 2.29× downside capture on 35/37 R2 trades — the regime classifier already exists and already ran, but its output didn't touch ranking.

TECHNICAL APPROACH

Context_Adjusted_Score = Base_Ensemble_Score + α · Sector_Regime_Score + β · Market_Regime_Score
Sector_Regime_Score $\in [-1, +1]$, derived from sector 20d relative strength + breadth + leadership concentration
 α, β fit via walk-forward optimization on trailing expectancy, not hand-tuned

```
def context_adjusted_rank(candidates, asof_date):
    sector_state = multi_layer.sector_regime_snapshot(asof_date) # already computed daily
    market_state = multi_layer.market_regime_snapshot(asof_date) # already computed daily
    scored = []
    for c in candidates:
        sector_adj = sector_state[c.sector].regime_score # STRONG=+1 .. STRESSED=-1
        market_adj = market_state.regime_score
        adjusted = c.base_score + ALPHA * sector_adj + BETA * market_adj
        scored.append((c.ticker, adjusted, c.base_score, sector_adj, market_adj))
    return sorted(scored, key=lambda x: -x[1])
```

VALIDATION PLAN

- Retrospective A/B: re-rank every historical day's candidate list with and without the adjustment, holding the rest of the pipeline fixed; compare realized win rate, expectancy, and max drawdown of the top-N under each ranking.
- Specifically test the WEAKENING-regime subset first, since that's the one case already flagged with real numbers — does the adjustment correctly demote or resize positions during WEAKENING days, and does that improve the 28.6% win rate / 2.29× downside capture figures already on record?
- Grid-search α, β on a training window, validate on a disjoint walk-forward window (see Part 4) to avoid overfitting two free parameters to noise.

Acceptance criteria: statistically significant expectancy improvement (paired bootstrap, Part 4 methodology) on out-of-sample walk-forward folds, with no significant degradation in NORMAL-regime performance (the adjustment shouldn't just move risk around).

P3 • Knowledge-Graph Community-Relative Scoring

Evidence: 581-node graph, 57 communities (Q=0.86), already computed via label propagation for DEV031 — currently feeds only the /why explainability trace.

TECHNICAL APPROACH

Community_Percentile(stock) = rank(ensemble_score of stock within its community) / |community|
Final_Score = (1- γ) · Global_Percentile + γ · Community_Percentile ($\gamma \in [0, 0.4]$, walk-forward tuned)

This directly answers the "Sector = Pharma is too coarse" problem raised earlier in this project: the 57 communities are a data-driven, emergent grouping (no hardcoded taxonomy), so a stock's community-relative rank captures peer comparison at a finer grain than the static Sector field.

```
def community_relative_rank(candidates, kg_snapshot):
    by_community = group_by(candidates, key=lambda c: kg_snapshot.community_of(c.ticker))
    for community_id, members in by_community.items():
        members_sorted = sorted(members, key=lambda c: c.base_score)
        n = len(members_sorted)
        for i, m in enumerate(members_sorted):
            m.community_percentile = i / max(n - 1, 1)
    for c in candidates:
        c.final_score = (1 - GAMMA) * c.global_percentile + GAMMA * c.community_percentile
    return candidates
```

SECONDARY USE — GRAPH INFLUENCE AS A FEATURE

The graph's existing personalized-PPR influence score (already computed) can be added as a plain feature-store column (kg_influence_score) and tested for predictive value the same way any other feature would be — via permutation importance against realized outcome, not assumed.

VALIDATION PLAN

- Backtest with $\gamma=0$ (baseline, current behavior) vs. γ swept over [0.1, 0.2, 0.3, 0.4] on walk-forward folds.
- Check community stability over time — if communities reshuffle heavily month to month, percentile ranks become noisy; measure community membership turnover as a data-quality precondition before trusting this signal.
- Cross-check against $\$P4$ (Cap×Sector) since community membership likely correlates with cap/sector — confirm this adds information beyond

what Cap×Sector already explains (see interaction-analysis methodology in Part 4).

Acceptance criteria: non-zero, statistically significant improvement in expectancy at the best γ , confirmed on an out-of-sample fold the γ sweep didn't touch.

P4 • Cap × Sector Interaction Study

Evidence: R2 LargeCap 100% win/+3.36% vs. MidCap 51.7%/+0.76% (n=6 vs. n=29); Technology 100%/+4.35% vs. Healthcare 0%/−1.13% (n=5 vs. n=3) — both independently reproduced, but not yet tested jointly for confounding.

TECHNICAL APPROACH

```
def interaction_table(outcome_dataset, dims=["cap", "sector"], min_n=5):
    groups = outcome_dataset.groupby(dims)
    rows = []
    for key, g in groups:
        n = len(g)
        tier = sample_tier(n)    # observation / hypothesis / research_signal / stronger_evidence / validation_candidate
        rows.append({
            "group": key, "n": n, "tier": tier,
            "win_rate": (g.pnl > 0).mean(),
            "avg_pnl": g.pnl.mean(), "median_pnl": g.pnl.median(),
            "profit_factor": g[g.pnl>0].pnl.sum() / abs(g[g.pnl<0].pnl.sum() or 1),
            "max_dd": g.max_dd.min(),
        })
    return pd.DataFrame(rows)

# Confound check: does Sector add information beyond Cap alone?
model_cap_only = fit_logistic(outcome_dataset, features=["cap"])
model_cap_sector = fit_logistic(outcome_dataset, features=["cap", "sector"])
likelihood_ratio_test(model_cap_only, model_cap_sector)    # is the addition significant?
```

VALIDATION PLAN

- Full interaction table across Runner × Cap × Sector × Investability, each cell tagged with its sample-size tier — no cell below n=5 gets more than an "observation" label, per the locked rule.
- Likelihood-ratio test (or equivalent) to determine whether Sector's apparent effect survives after controlling for Cap, and vice versa — directly resolves the "is Technology's edge really Technology, or just LargeCap/liquidity" question raised earlier.

Acceptance criteria: this is a research deliverable, not a pass/fail gate — its output (which interaction cells show real, non-confounded signal) is what determines whether P2/P3's adjustment terms should include a Cap component, a Sector component, or both.

P5 • Remaining R2 Upgrades (grouped, shorter form)

Upgrade	Formula / approach	Validation
Ensemble disagreement display	disagreement = stdev(model_1..11 scores), normalized to [0,1] across the day's candidate set	Correlate disagreement with [actual return − predicted return]; if correlated, promote to a sizing multiplier (size = base_size * (1 - disagreement)) after P1's calibration gate clears.
Regime-conditional ensemble weights	Maintain w_i^{regime} per model per regime bucket, computed from IC restricted to days in that bucket; fall back to global weights if bucket $n < 30$ trading days	Backtest regime-conditional vs. global weights on walk-forward folds, segmented by regime.
Daily turnover / rotation cap	Cap total rotation-driven reallocation to X% of NAV/day; when signals exceed budget, execute top-N by expected-alpha-delta, defer the rest	Compare realized cost (est. slippage × turnover) before/after cap on a simulated replay of historical rotation days.
Point-in-time universe audit	Reconstruct NSE 200 / S&P 500+MidCap 400 membership as of each historical trade date; re-verify the 539/1,060-trade evidence base	One-time audit; if membership drift is found, all backtest metrics cited in this project need a footnoted correction.
Standing post-R1 comparator	Fixed, un-optimized baseline (e.g. equal-weight top-10 by 3-month momentum, monthly rebalance) run permanently in parallel paper mode	No tuning allowed on this baseline, ever — its entire value is being a fixed yardstick, not a competitor to optimize.

Part R3 · Runner 3 — Deep Technique Specifications

Runner 3's research question, sharpened: *can context + regime + institutional flow + calibrated uncertainty produce better risk-adjusted stock selection than R2* — not "can we build a more complicated model." Every technique below is tiered (Tier 1 = build now, Tier 2 = evidence-gated add, Tier 3 = deferred) consistent with the priorities already locked.

II.1 · Model Families

GRADIENT-BOOSTED TREES (TIER 1 — PRIMARY MODEL FAMILY)

```
# Walk-forward training loop (see Part 4 for fold structure)
for fold in walk_forward_folds(data, train_window=252, test_window=63, step=21, embargo=5):
    model = LightGBM(
        objective="binary", metric="auc",
        num_leaves=31, max_depth=6, learning_rate=0.03,
        min_data_in_leaf=50,          # guards against overfitting on thin feature cells
        feature_fraction=0.7, bagging_fraction=0.7, bagging_freq=5,
        lambda_l1=1.0, lambda_l2=1.0,
    )
    model.fit(fold.train_X, fold.train_y, early_stopping_rounds=50,
              eval_set=[(fold.val_X, fold.val_y)])
    calibrated = fit_platt(model.predict(fold.val_X), fold.val_y)
    fold_result = evaluate(model, calibrated, fold.test_X, fold.test_y)
    log_fold_result(fold_result)
```

Feature importance via SHAP values (not raw gain, which is biased toward high-cardinality features) — ranked and reviewed every retraining cycle, with any feature whose mean |SHAP| falls below a floor for 3 consecutive cycles flagged for removal (feature decay monitoring).

ENSEMBLE STACKING (TIER 2)

```
Meta-learner: P_stacked = sigmoid(w1·P_gbm + w2·P_r2_base + w3·P_kg_community + b), weights fit via held-out logistic regression,
                                         never hand-set
```

Only pursue once each base model (GBM, R2's existing base score, KG community score) has independently demonstrated standalone signal — stacking noise on noise doesn't create signal.

GRAPH NEURAL NETWORK ON THE KNOWLEDGE GRAPH (TIER 2/3 — GENUINELY NOVEL, HIGHER ENGINEERING COST)

Deeper version of P3's community-relative scoring: instead of a static percentile within a community, learn a representation via message-passing over the graph directly.

```
# GraphSAGE-style sketch — 2-layer message passing
h0 = node_features(stock, sector, fundamentals, momentum)      # initial embedding
for layer in [1, 2]:
    h_neighbors = mean_aggregate([h[layer-1][n] for n in graph.neighbors(stock)])
    h[layer][stock] = relu(W[layer] @ concat(h[layer-1][stock], h_neighbors))
final_embedding = h[2][stock]
score = linear_head(final_embedding)          # trained end-to-end against realized outcome
```

Risk (explicitly acknowledged, per the platform's own caution on deep models for financial tabular data): 581 nodes is a small graph by GNN standards, and overfitting risk is real. Treat this as a Tier 3 stretch experiment gated behind Tier 1/2 GBM+community-percentile results actually working first — don't build the GNN as a first move.

BAYESIAN MODEL AVERAGING (TIER 2)

```
Posterior_weight(model_i) ∝ Prior(model_i) · Likelihood(model_i's historical IC | data)
Posterior_predictive(stock) = Σ_i Posterior_weight(model_i) · P_i(stock)
→ produces a genuine predictive distribution, not one point score
```

Directly enables the "Expected Return: median / range / confidence" output format this project flagged as a goal much earlier — a natural replacement for a bare point "Expected Alpha %" figure once implemented.

II.2 · Relative-Value & Market-Neutral Construction

FACTOR-NEUTRAL SCORING (TIER 2)

```
Residual_Score(stock) = Base_Score - Σ_k β_k · Factor_Exposure_k(stock)
β_k estimated via cross-sectional regression of Base_Score on [size, value, momentum] factor exposures, each trading day
```

Tests whether R2/R3's apparent LargeCap/Technology edge is genuine stock-picking skill or an unintentional, uncompensated factor tilt — directly extends the P4 interaction study.

PEER-PAIR STATISTICAL ARBITRAGE (TIER 3)

```
Cointegration test (Engle-Granger): regress price_A on price_B → residual spread; ADF test on residual for stationarity (p < 0.05)
Entry: |spread z-score| > 2.0      Exit: |spread z-score| < 0.5      Stop: |spread z-score| > 3.5
```

Candidate pairs sourced from the same Knowledge-Graph community (P3) rather than a blind universe-wide search — cheaper to screen, more economically motivated pairing than pure statistical mining. Requires short-selling infrastructure this platform doesn't currently have; this is the single biggest engineering lift on the whole list and should stay Tier 3 until Tier 1/2 work is validated.

II.3 · Regime & Change-Point Detection

CUSUM-BASED REGIME SHIFT DETECTION (TIER 3, SUPPLEMENTS THE EXISTING 5-STATE CLASSIFIER)

$$S_t = \max(0, S_{(t-1)} + (x_t - \mu_0 - k))$$

Flag regime shift when $S_t > h$ (h, k tuned on historical regime-transition dates as ground truth)

Where x_t is a rolling market-breadth or volatility statistic. This is a more principled, continuously-updating alternative to discrete NORMAL/WEAKENING/RISK_OFF/CRASH/RECOVERY buckets, useful for catching a regime *transition in progress* rather than only classifying the current state after the fact. Test as a leading indicator against the existing classifier's own bucket transitions before trusting it standalone.

II.4 · Quality & Red-Flag Screens (named, auditable formulas)

PIOTROSKI F-SCORE (TIER 1 — 9 BINARY CRITERIA, SUM = 0-9)

#	Criterion	Pass condition
1	Positive net income	ROA > 0 this year
2	Positive operating cash flow	CFO > 0
3	Improving ROA	ROA(t) > ROA(t−1)
4	Quality of earnings	CFO > Net Income (accrual check)
5	Lower leverage	Long-term debt/assets decreasing YoY
6	Improving liquidity	Current ratio increasing YoY
7	No dilution	Shares outstanding not increased YoY
8	Improving margin	Gross margin increasing YoY
9	Improving asset turnover	Asset turnover ratio increasing YoY

BENEISH M-SCORE (TIER 1 — EARNINGS-MANIPULATION SCREEN, 8-VARIABLE)

$$M = -4.84 + 0.92 \cdot DSRI + 0.528 \cdot GMI + 0.404 \cdot AQI + 0.892 \cdot SGI + 0.115 \cdot DEPI - 0.172 \cdot SGAI + 4.679 \cdot TATA - 0.327 \cdot LVGI$$

$M > -1.78 \rightarrow$ elevated manipulation-risk flag (standard threshold)

Both scores are computable from fundamentals data already ingested — this is a zero-new-data-source, Tier 1 addition. Replacing the current general "Quality" composite with these named, published, auditable scores is strictly more explainable than the current black-box quality factor, and directly serves the "picking the right stocks is mandatory" goal by catching accounting red flags a pure price/momentum/fundamentals-ratio screen would miss.

GOVERNANCE/PROMOTER SCREEN — INDIA-SPECIFIC (TIER 2)

Pledged-share percentage (from shareholding-pattern filings) and related-party-transaction flag frequency — known, India-specific sources of midcap/smallcap blowups not captured by standard fundamental ratios.

II.5 · NLP Depth & Estimate-Revision Signals

ANALYST ESTIMATE REVISION MOMENTUM (TIER 1)

$$\text{Revision_Momentum} = (\text{Consensus_EPS_now} - \text{Consensus_EPS_3mo_ago}) / |\text{Consensus_EPS_3mo_ago}|$$

A well-established, cheap factor distinct from the level of fundamentals already used — captures direction of change in Street expectations, not just where they currently sit.

EARNINGS CALL TRANSCRIPT TONE (TIER 2)

Beyond headline sentiment (already covered by the existing FinBERT news pipeline): apply a finance-tuned sentiment/tone classifier to transcript segments, track hedging-language frequency ("uncertain," "challenging," "headwinds") quarter-over-quarter as a distinct signal from what's said in the prepared remarks vs. the Q&A section specifically (Q&A tone is generally considered more informative than prepared remarks in the literature, since it's less scripted).

II.6 · Multi-Horizon Consensus

Important boundary, restated: this is a feature inside R3, not a merge with the separate R004 intraday product. R004 keeps its own P&L, lifecycle, and infrastructure; this technique borrows only the idea of a shorter-horizon confirmation signal.

$$\text{Agreement_Score} = \text{sign_match}(\text{Prediction_5day}, \text{Prediction_17day}) \cdot \min(|\text{Prediction_5day}|, |\text{Prediction_17day}|)$$

High agreement → conviction boost; disagreement → treat as a WAIT signal rather than a rejection

```
def multi_horizon_check(stock, model_5d, model_17d):
    p5 = model_5d.predict(stock.features)
    p17 = model_17d.predict(stock.features)
    if sign(p5) == sign(p17):
        return {"verdict": "CONFIRMED", "conviction_multiplier": 1.15}
    else:
        return {"verdict": "MIXED_TIMING", "conviction_multiplier": 0.7}
```

Both horizon models share the same underlying feature base (Category L design) — this is meant to be cheap to add once the Tier 1 GBM model exists, not a second full pipeline.

III.1 • Walk-Forward Protocol (applies to every item in Parts I and II)

```
def walk_forward_folds(data, train_window=252, test_window=63, step=21, embargo=5):
    """
    train_window: ~1 trading year of history to fit on
    test_window: ~1 quarter held out, never seen during fit
    step:        slide forward ~1 month per fold
    embargo:      5-day gap between train and test to prevent
                  feature leakage from overlapping horizons
                  (e.g. a 17-day holding period signal computed
                  near the train/test boundary can leak into the
                  test window without an embargo)

    """
    folds = []
    start = data.first_date
    while start + train_window + embargo + test_window <= data.last_date:
        train = data[start : start+train_window]
        test  = data[start+train_window+embargo : start+train_window+embargo+test_window]
        folds.append(Fold(train, test))
        start += step
    return folds
```

This embargo detail matters and is easy to skip: any signal derived from a forward-looking holding-period outcome (which several R2/R3 features implicitly are, since "did this trade eventually win" is part of the training signal) can leak across a train/test boundary that sits inside the holding window. A 5-trading-day embargo is a minimum, not a formality.

III.2 • Statistical Testing

Test	When to use	Note
Paired bootstrap (10,000 resamples)	Comparing two strategies/rankings on the same historical positions (e.g. P0's actual vs. counterfactual exits, P2's adjusted vs. unadjusted ranking)	Preferred over a t-test here because trade returns are fat-tailed and non-normal — bootstrap makes no distributional assumption.
Likelihood-ratio test	Does adding a variable (e.g. Sector, on top of Cap) significantly improve a model, per P4	Standard nested-model comparison; cheap and appropriate for the Cap×Sector confound question specifically.
Deflated Sharpe Ratio / White's Reality Check	Whenever multiple strategies or parameter combinations are tested and the best one is being selected (this applies to nearly everything in Part II, since a large catalog of techniques is being screened)	This is the most important, most commonly skipped test in this whole document. Testing 15+ techniques and picking whichever backtest looks best is a multiple-comparisons problem — some will look good by chance alone. The Deflated Sharpe Ratio explicitly corrects the apparent Sharpe ratio for the number of trials run before selecting the winner.

$$\text{Deflated Sharpe} \approx \text{Sharpe}_{\text{observed}} - (\text{expected max Sharpe under } N \text{ trials with no real skill})$$
$$\text{Expected max Sharpe under } N \text{ trials} \approx \sigma_{\text{Sharpe}} \cdot [(1-\gamma) \cdot \Phi^{-1}(1-1/N) + \gamma \cdot \Phi^{-1}(1-1/(N \cdot e))]$$
$$(\gamma = \text{Euler-Mascheroni constant} \approx 0.5772; N = \text{number of strategy variants trialed})$$

Concretely for this project: if the Runner 3 catalog tests, say, 12 ranking-adjustment variants (different $\alpha/\beta/\gamma$ combinations across P2/P3/factor-neutral construction) and picks the best one, that "best" result must be deflated for having run 12 trials before reporting a Sharpe ratio — otherwise the reported number is optimistic by construction, not by merit.

III.3 • Experiment Matrix Template

Experiment	Hypothesis	Method	Metric	N trials in family (for deflation)	Status
P0 exit-bridge replay	Disciplined exits improve expectancy, esp. in WEAKENING/RISK_OFF	Walk-forward counterfactual replay, n=539	Expectancy delta, paired bootstrap CI	1 (single mechanism, not swept)	Planned
P1 confidence calibration	Calibrated confidence is monotonic with outcome; raw is not	Weekly reliability diagram, rolling 180d	ECE, Brier score	1	Planned
P2 α/β sweep	Sector/market regime adjustment improves expectancy	Walk-forward grid search + holdout	Expectancy, max DD, Sharpe (deflated)	≈ 9 (3×3 grid)	Planned
P3 γ sweep	Community-relative scoring adds signal beyond global rank	Walk-forward grid search + holdout	Expectancy (deflated for 4-5 γ values)	≈ 5	Planned
R3 GBM baseline	Tier-1 features on GBM beat R2 base rank	60-90d live shadow vs. R2	Sharpe, win rate, expectancy (deflated)	1 (single model family at this stage)	Blocked on P0 Outcome Dataset

Part 5 · Phased Implementation Roadmap

Phase	Scope	Duration (est.)	Gate to proceed
Phase 0	P0 Outcome Dataset + data dictionary finalized (already locked as prerequisite); PIT universe audit; exit-bridge retrospective replay	2–3 weeks	Outcome Dataset queryable by Position ID; exit-bridge replay statistically conclusive (n≥50)
Phase 0.5	R2 zero-entry diagnosis (parallel with Phase 0, gates Phase 0.5's own advisory ship); R1 advisory sheet (Stage 1, Part R1 §R1.9)	1 week	Preliminary R2 diagnosis read exists before R1 advisory ships; banner explicitly states R1 positions carry no dynamic-exit protection
Phase 1	P1 confidence calibration wired to delivered output; P4 Cap×Sector interaction study completed	2–3 weeks	ECE ≤0.05 sustained 4 weeks; interaction table complete with confound test results
Phase 2	P2 sector/regime ranking adjustment (backtest + walk-forward validation); P3 KG community-relative scoring (backtest)	3–4 weeks	Both clear deflated-Sharpe-corrected significance on out-of-sample folds
Phase 2 (parallel)	R3 Tier 1 build: existing daily features + FII/DII + earnings calendar + options PCR + GBM + Platt calibration, fully isolated per Part 0 contract	3–4 weeks	R3 isolation CI checks pass; R3 replicates R2 baseline before adding new features
Phase 3	P5 remaining R2 upgrades (disagreement display, regime-conditional weights, turnover cap, standing comparator); R3 begins 60-90 day live shadow vs. R2	ongoing / 60–90 days for shadow	Day-30 kill gate on R3 shadow before continuing to Day 90
Phase 4	R3 Tier 2 additions only where Phase 3 shadow results justify them (per-technique evidence gate, not a batch add)	evidence-driven, no fixed date	Each Tier 2 addition requires its own Research Ticket per the locked governance
Phase 5	Promotion decision: does anything from R3 clear the bar to become a CEO-authorized R2 upgrade	evidence-driven	Explicit CEO authorization, same pattern as R1 retirement / Phase 2 identity migration — never automatic

Phase 2's two tracks (R2 upgrades and R3 Tier 1 build) run in parallel deliberately — they don't block each other, and neither should be delayed waiting on the other, since the isolation contract in Part 0 means they can't interfere even while running simultaneously.

Part 6 · Governance Recap & Acceptance Gates

- **Sample-size tiers** (locked earlier in this project, applied throughout this document): n<5 observation only, 5–14 hypothesis, 15–29 research signal, 30–49 stronger evidence, 50+ validation candidate.
- **Multiple-testing correction is now an explicit requirement**, not implicit — every experiment family in the matrix above records its trial count so reported Sharpe/expectancy figures can be deflated before anyone treats them as final.
- **No R2 production change without walk-forward validation + statistical significance**, per the acceptance criteria stated under each item in Part R2.
- **No R3 result reaches R2 without explicit, named, dated CEO authorization** — the isolation contract in Part 0 makes this structurally enforced, not just procedurally requested.
- **The Outcome Dataset (P0, already locked) is the required foundation** for nearly everything in this document — Phase 0 above is not optional preamble, it's the dependency the rest of the roadmap sits on.
- **Eighth Research Trigger — Signal Silence** (Part R1, §R1.7): a runner producing zero new positions for N≥10 trading days, while a comparator shows eligible setups, is now a formal, evidence-gated finding — not a fixed calendar-date advisory decision.
- **Minimum Viable Signal** (Part R1, §R1.7): daily cross-runner throughput is now itself a monitored metric, with a pre-registered, walk-forward-validated (never silent, never ad hoc) gate-relaxation policy if it falls below a stated floor.
- **No composite-layer output (R1×R2×R3) reaches an operator as an actionable size recommendation without clearing the same deflated-Sharpe-corrected, out-of-sample validation as everything else** — the 9× filter-improvement number in Part R1 is a hypothesis, stated as one, not a shipped rule.

Bottom line

This document turns the earlier catalog into something buildable across all three runners: every upgrade has a formula or pseudocode sketch, a validation method, and an acceptance gate, and every research technique is tiered by evidence-readiness rather than presented as one undifferentiated list. The three-track structure (R1 defensive specialist, R2 all-weather core, R3 research edge, running in parallel under one isolation contract) is the concrete answer to "keep the runners separate without disturbing any of them" — not a policy statement, but a set of CI-enforced boundaries none of the three can cross without a human decision, plus a genuine reason for three runners to exist together: differentiated mandates, a principled composite layer reusing R2's own proven math, and a structural answer to the single biggest operational risk raised in this round — that daily recommendation throughput, not just any one runner's win rate, is what keeps the whole evidence-gated research program fed.